Deep Learning for Building Change Detection in Multi-Temporal Satellite Imagery

Project proposal | DATA 255 Deep Learning Vision | Fall 2026

A. Title and Team

Team: BuildSafe Vision.
Members: Sneha Singh, Ritika Mukesh Neema, Danish Waseem, Rajesh Paruchuri.
Project area: Binary semantic segmentation, multi-temporal remote sensing, and efficient computer vision.

B. Problem and Research Question

Urban planners and infrastructure analysts need to identify where buildings have appeared or disappeared between satellite acquisitions. Direct image subtraction is unreliable because sunlight, vegetation, and acquisition conditions can change even when the built environment does not. We will study a paired-image segmentation system that predicts a binary map of *building-related changes*, rather than all visible scene differences. This is an academic research prototype, not an operational monitoring or damage-assessment system.

Research question. Under a matched training and evaluation protocol, does a shared-encoder Siamese U-Net improve building-change F1 over a conventional six-channel U-Net, and how do loss design and paired-image augmentation affect performance and computational cost?

Hypotheses. (H1) Siamese feature extraction will improve change-class F1 or IoU by explicitly comparing before/after representations, although it may require more inference time. (H2) Binary cross-entropy (BCE) plus Dice loss will improve changed-pixel detection relative to BCE alone because changed pixels are uncommon. Success means producing credible, repeatable evidence for or against these claims, not promising a particular score.

C. Dataset and Evaluation

LEVIR-CD [1] contains **637** aligned RGB satellite-image pairs, each 1024×1024 pixels, with expert-annotated binary masks for building construction or removal. The images come from 20 regions in Texas across different dates and conditions. Dataset access and local extraction have been verified. We use the available split of **445 training, 64 validation, and 128 test pairs**. The source images and labels are available at https://justchenhao.github.io/LEVIR/. Use is restricted to academic purposes and must respect the Google Earth terms.

We partition each 1024×1024 image into sixteen non-overlapping 256×256 paired tiles *after* assigning its source pair to a split. Thus, training has 7,120 tiles and validation has 1,024; the 128 test pairs stay isolated until final evaluation. Augmentation is applied identically to both images and the corresponding mask. The audit verified paired filenames and dimensions; it also identified a small number of nonbinary grayscale mask values, so masks will be binarized consistently at intensity ≥ 128 and unusual masks reviewed. We will audit available geospatial metadata for nearby or overlapping scenes and report residual geographic-leakage risk.

Primary metric: changed-class pixel F1 (Dice), which avoids the misleadingly high accuracy of predicting the dominant unchanged class. **Secondary metrics:** changed-class IoU, precision, recall, and per-image error summaries. We will aggregate pixel counts across the validation set and apply a predeclared decision threshold, with any tuning limited to validation. All settings and thresholds will be frozen before a single final test evaluation. Failure examples will include missed small buildings, false positives, and imprecise boundaries.

D. Related Work and Technical Approach

Ronneberger *et al.* [2] introduced U-Net's encoder-decoder design with skip connections, motivating a compact pixelwise segmentation baseline. Daudt *et al.* [3] explored fully convolutional Siamese networks for paired-image change detection, motivating shared feature extraction and comparison. Chen and Shi [1] introduced LEVIR-CD and STANet, demonstrating the importance of temporal relationships and nuisance variations; we will not claim to reproduce their attention architecture.

Experimental role	Method and purpose
Reference baseline	Absolute RGB image difference with a validation-selected threshold; a nonlearned control.
Strong baseline	Lightweight U-Net taking the two RGB images concatenated into six channels, predicting one change mask.
Team investigation	Shared-weight Siamese U-Net processing each date in a separate encoder branch, fusing learned features and decoding a change mask.

Models will be written in PyTorch using standard convolutional blocks; pretrained components are not required for the compact baseline. We will cite and attribute the U-Net, Siamese, dataset, and any reused code. The student-owned contribution is the controlled implementation, consistent evaluation, loss and augmentation ablations, reliability study, and failure analysis.

E. Experimental Plan

The **principal comparison** is six-channel U-Net versus Siamese U-Net using the same training/validation pairs, 256×256 tiles, augmentation policy, BCE+Dice loss, optimizer, learning rate, batch-size rule, maximum epoch budget, and validation-F1 checkpoint selection. We will document parameter counts and any unavoidable architectural differences. The models will use identical evaluation code and mask thresholding.

1. **Ablation 1, loss:** train the selected architecture with BCE only versus BCE+Dice, holding all other settings fixed. Report F1, IoU, precision, and recall to detect false-positive trade-offs.
2. **Ablation 2, paired augmentation:** compare synchronized random flips and right-angle rotations against no geometric augmentation at the same resolution, architecture, and training budget.
3. **Computational trade-off:** measure training time, parameter count, checkpoint size, and batch-one inference latency for both models. Timing will include a declared preprocessing convention, warm-up, and device synchronization. We will compare quality per unit of computation, not just peak F1.

The main trained-method comparison targets **three seeds** (1511, 42, 123), with mean, standard deviation, and individual results reported. Ablations can initially use one seed, followed by confirmation if the observed difference is meaningful and compute allows. We will preserve run configurations, software versions, curves, masks, and qualitative examples. Model selection and ablation decisions use validation only. A separate analysis of difficult scenes will not be used to retune the final test result.

F. Feasibility and Team Plan

The complete dataset has already been downloaded and audited. A 30-epoch six-channel U-Net run completed on an **Apple M5 with 16 GB unified memory using PyTorch MPS**; it achieved a **preliminary validation F1 of 0.7319** (IoU 0.5772), with the best checkpoint at epoch 24. This is a single-seed feasibility result, not a locked-test score or an estimate of model superiority. The run took about **2.7 hours**; we will benchmark Siamese training before fixing the total schedule. We reserve roughly **25–40 accelerator-hours** for principal runs, ablations, and reruns, using the university GPU lab if local training becomes slow.

Planned milestones. Nov. 2–8: finalize data/metadata checks and reproducibility setup. Nov. 9–22: implement and benchmark Siamese U-Net. Nov. 23–Dec. 6: complete loss and augmentation ablations and additional seeds. Dec. 7–11: freeze choices, evaluate the locked test set once, and analyze errors and efficiency. Dec. 12–16: cross-member reproduction, report, presentation, and final submission.

Proposed responsibilities. Sneha: dataset integrity, geospatial overlap checks, and reference-difference baseline. Ritika: U-Net training pipeline, controlled augmentation, and seed reproducibility. Danish: Siamese U-Net implementation, loss comparison, and error analysis. Rajesh: validation metrics, efficiency benchmarks, statistical summaries, and reproducibility verification. Each member will contribute code and writing, review another component, and be able to explain the full experiment.

Main risk and fallback. Geographic similarity across splits, class imbalance, and training cost may limit generalization claims. Our *minimum viable project* is a reproducible reference method, U-Net versus Siamese U-Net comparison, two single-seed ablations, and fully documented validation analysis. If three-seed training is infeasible, we will prioritize repeated runs for the main comparison and request instructor approval for a reduced variability plan. A 512-pixel tile study is a stretch goal only.

References

[1] H. Chen and Z. Shi, "A Spatial-Temporal Attention-Based Method and a New Dataset for Remote Sensing Image Change Detection," *Remote Sensing*, vol. 12, no. 10, art. 1662, 2020. DOI: 10.3390/rs12101662. Dataset: https://justchenhao.github.io/LEVIR/.

[2] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in *MICCAI*, pp. 234–241, 2015. DOI: 10.1007/978-3-319-24574-4_28.

[3] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully Convolutional Siamese Networks for Change Detection," in *IEEE ICIP*, pp. 4063–4067, 2018. DOI: 10.1109/ICIP.2018.8451652.

[4] H. Chen and Z. Shi, *STANet*, public research implementation. https://github.com/justchenhao/STANet. Accessed November 2026.

Assistance Disclosure

OpenAI ChatGPT was used strictly as a reference tool to assist with formatting, light structural organization, and checking bibliographic details during the preparation of this proposal.

All research questions, experimental methodologies, feasibility assumptions, and responsibility assignments were developed independently by the team. All reported model training, dataset audit results, and empirical measurements reflect direct output from the team's executed experiments.